

Ahsanullah University of Science and Technology



Department of Computer Science and Engineering

Malaria Disease Detection Using Deep Neural Network

Submitted To

Mr. Mir Tafseer Nayeem

Assistant Professor

Dept. of CSE, AUST

Submitted By:

Name: Fakhrul Islam Fahad

Id: 16.02.04.051

Email: 160204051@aust.edu

Name: Dipesh Shome

Id: 16.02.04.045

Email: 160204045@aust.edu

Name: Arnab Saha

Id: 16.02.04.036

Email: 160204036@aust.edu

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Introduction:

- ❖ **Definition of the Task:** Malaria is a serious and sometimes fatal disease caused by a parasite that commonly infects a certain type of mosquito which feeds on humans. People who get malaria are typically very sick with high fevers, shaking chills, and flu-like illness. Four kinds of malaria parasites infect humans: *Plasmodium falciparum*, *P. vivax*, *P. ovale*, and *P. malariae*. In addition, *P. knowlesi*, a type of malaria that naturally infects macaques in Southeast Asia, also infects humans, causing malaria that is transmitted from animal to human (“zoonotic” malaria). *P. falciparum* is the type of malaria that is most likely to result in severe infections and if not promptly treated, may lead to death. Although malaria can be a deadly disease, illness and death from malaria can usually be prevented.
- ❖ **Motivation:** According to The World Health Organization (WHO), almost half of the world’s population is at the risk of malaria. Over 219 million malaria cases had been reported in 2019 worldwide and 435,000 deaths in that year. Bangladesh also suffers significantly because of malaria. The southeast areas of the Chittagong Hill Tract were declared as a malaria hotspot. The northeast districts of Bangladesh, namely Sunamganj and Netrokona, were the hotspot for 2013 to 2016. The disease is regularly found in 13 of 64 areas, with over 8% of the total population (Over 17 million people) at risk. Predominantly malaria is a highly seasonal disease in Bangladesh. Mainly in the hills and rural areas of Bangladesh, there is a shortage of expert technicians or medical professionals who can classify whether the blood sample is infected or not. So, when the infected number of people spiked in the past, many people died because of that. But early detection could have saved many lives. So, our motivation is to detect malaria rapidly with improved accuracy and reduce the workload of human experts.
- ❖ **Applications:** There are many applications that rely on deep neural networks. We can use it on natural Language Processing, colourisation of Black and White images, automatic speech recognition, image recognition etc.
- ❖ **Importance:** Malaria becomes a global health problem in Asian countries. It distributes comprehensively in the tropical and subtropical climate of the region. According to the World Health Organization (WHO), 300-500 million people are infected with malaria every year, and more than one million people die. Most of these deaths occurred in the African region (92%). Millions of people from all around the world are still lacking malaria prevention and treatment. For that reason, nearly half of the world population is at the risk of malaria [WHO, 2016]. So, it’s important to detect this life threatening disease using advanced technology.

We choose a deep neural network to detect malaria disease. Deep neural networks have recently become the standard tool for solving a variety of problems. Deep neural networks offer a lot of value to statisticians, particularly in increasing accuracy of a machine learning model.

- ❖ **Challenges:** This dataset is a highly non linear dataset so that the model accuracy is not too high. Furthermore during model creation we face some issues. To solve these problems we google the errors but all the solutions are for convolution neural networks So we spent a huge time on bug solutions.

Related Works:

- ❖ **List of Some Works:** There are some works related to Malaria detection.
 - Improving Malaria Parasite Detection from Red Blood Cell using Deep Convolutional Neural Networks.
 - Malaria Detection Using Deep Residual Networks with Mobile Microscopy.
 - Classification of Malaria Infected Cells Using Deep Convolutional Neural Networks.
 - Malaria Detection using Deep Learning.
 - CNN-Based Image Analysis for Malaria Diagnosis.

- ❖ **Summarization:**

One article focuses on improving malaria detection from patches segmented from microscopic images of red blood cell smears by introducing a deep convolutional neural network. Compared to the traditional methods that use tedious hand engineering feature extraction, the proposed method uses deep learning in an end-to-end arrangement that performs both feature extraction and classification directly from the raw segmented patches of the red blood smears. The dataset used in this study was taken from the National Institute of Health named NIH Malaria Dataset. The evaluation metric accuracy and loss along with 5-fold cross validation was used to compare and select the best performing architecture. To maximize the performance, existing standard pre-processing techniques from the literature have also been experimented. In addition, several other complex architectures have been implemented and tested to pick the best performing model. A holdout test has also been conducted to verify how well the proposed model generalizes on unseen data. The best model achieves an accuracy of almost 97.77%.

Another work shows the traditional way of diagnosing malaria is by schematic examining blood smears of human beings for parasite-infected red blood cells

under the microscope by lab or qualified technicians. This process is inefficient and the diagnosis depends on the experience and well knowledgeable person needed for the examination. Deep Learning algorithms have been applied to malaria blood smears for diagnosis before. However, practical performance has not been sufficient so far. This paper proposes a new and highly robust machine learning model based on a convolutional neural network (CNN) which automatically classifies and predicts infected cells in thin blood smears on standard microscope slides. A ten-fold cross-validation layer of the convolutional neural network on 27,558 single-cell images is used to understand the parameter of the cell. Three types of CNN models are compared based on their accuracy and select the precise accurate - Basic CNN, VGG-19 Frozen CNN, and VGG-19 Fine Tuned CNN. Then by comparing the accuracy of the three models, the model with a higher rate of accuracy is acquired.

Project Objectives:

- ❖ **Tasks of the System:** The main objective of this project is to design a binary class classification model that can recognise Malaria infected and uninfected cells from Malaria cell images dataset. This dataset contains 27558 images. We will try to implement this project using Deep Neural Network with different hyperparameter settings in pytorch.

We characterize the complete work flow into nine steps.

- 1) Preprocessing of CSV data and corresponding image data.
- 2) Normalize the input data.
- 3) Train-Test split.
- 4) Setting up hyperparameters.
- 5) Initialize model, loss and optimizer.
- 6) Training data and evaluation on validation data in terms of loss and accuracy.
- 7) Save Model.
- 8) Build Binary class classification model
- 9) Prediction from test dataset.

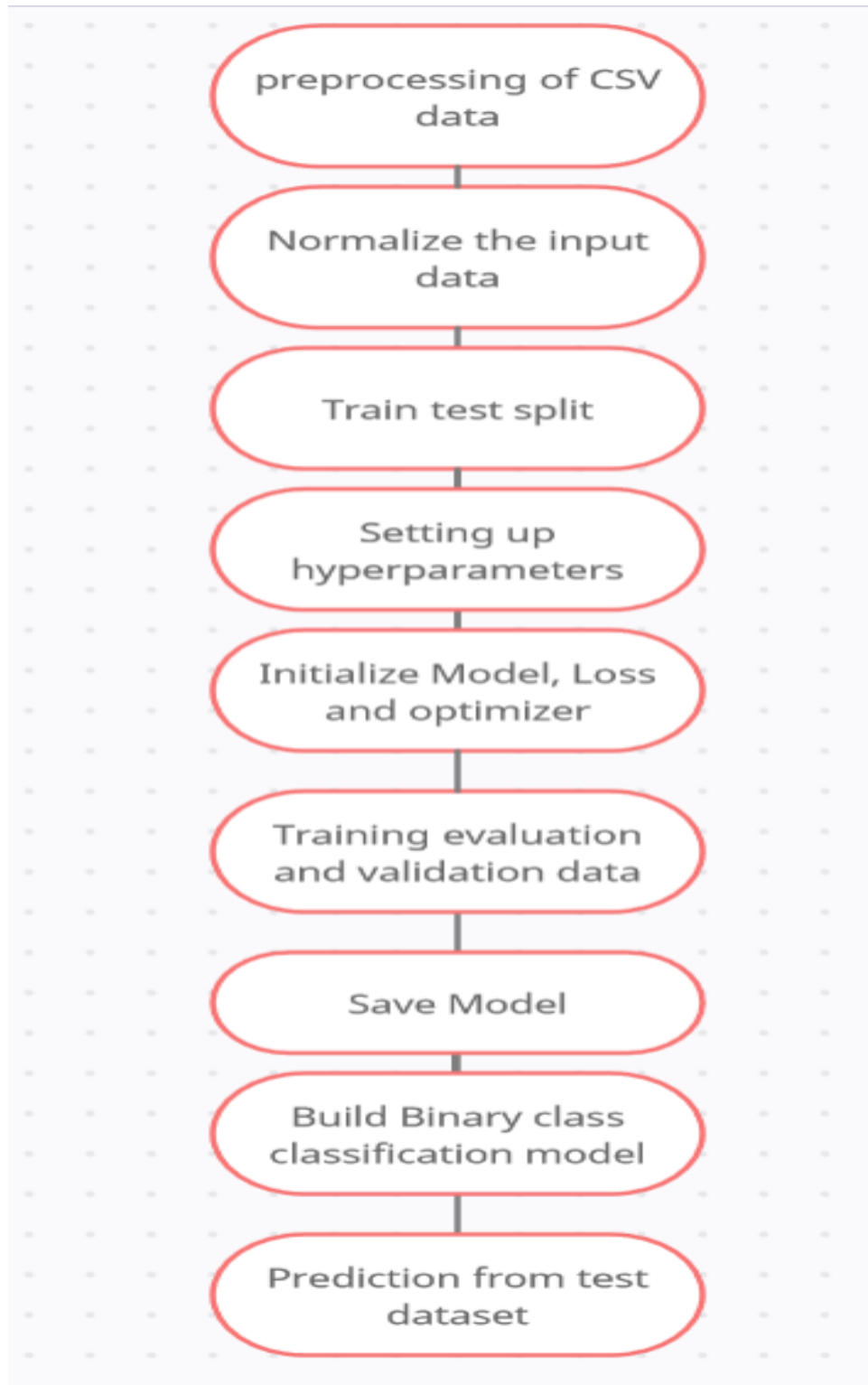
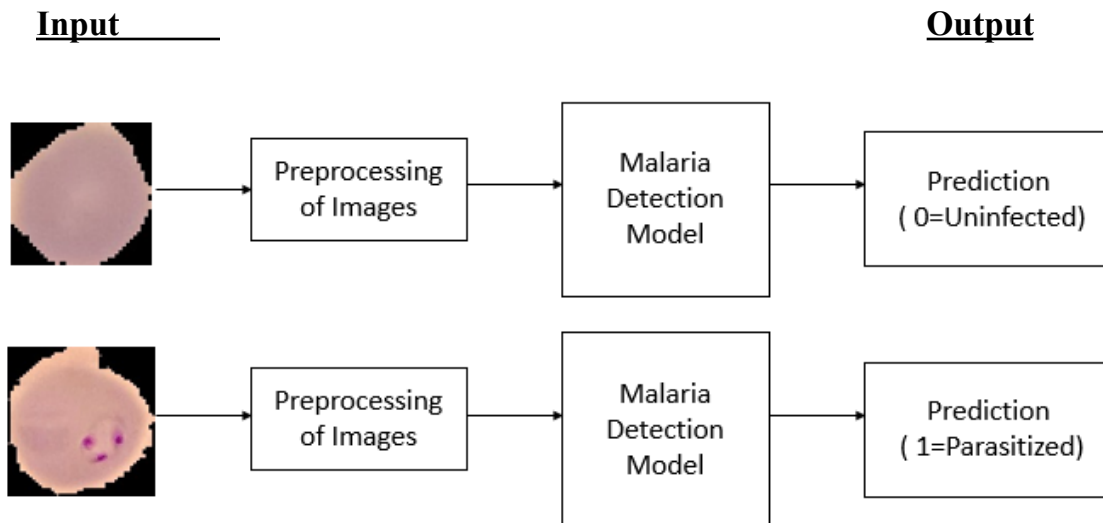


Figure: Flowchart of the work

❖ Dummy Input and Output:



Methodologies / Model:

❖ **Solve of the Problem:** Our main objective is to make an automated diagnosis system for malaria cell detection using deep neural networks. First we followed some preprocessing steps in input data. Then we created models consisting of an input layer, three hidden layers with relu or leakyrelu activations and output layer. During building our model, we vary the number of hidden units in the hidden layer with different hyperparameters on different models. Briefly discussed below:

- **Preprocessing:** This dataset is a combination of 2 folders. Then we created a CSV file for the corresponding folder data. For training we have to merge the training image dataset and then merge the csv dataset. After that we process the image with a csv dataset to attach images with labels. We apply normalization using pytorch transform function which transform the image into tensor form and resize in 28*28 and then apply normalize function to reduce computing cost during training. $\text{image} = (\text{image} - \text{mean}) / \text{s.deviation}$ where the value of mean and standard deviation is 0.5.
- **Models:** For our task, we created three different models. Each model consists of different hidden layers with different activation functions. Given below:

Model 1	Model 2	Model 3
Hidden layer=3	Hidden layer=3	Hidden layer=3
Hidden unit: 512, 256, 128	1024, 560 256	1000, 600 ,200
Activations: Relu, leakyReLU	Activations:Tanh, Relu, leakyReLU	Activations:Tanh, Relu, leakyReLU
Optimizer: SGD	Optimizer: Adam	Optimizer: Adam
Loss: CrossEntropy	Loss: CrossEntropy	Loss: CrossEntropy

ReLU: A Rectified Linear Unit (A unit employing the rectifier is also called a rectified linear unit ReLU) has output 0 if the input is less than 0, and raw output otherwise. That is, if the input is greater than 0, the output is equal to the input.

LeakyReLU: When a large number of dead neurons are there in a Neural Network it's performance is affected, this can be corrected by making use of what is called Leaky ReLU where the slope is changed .

Tanh: The hyperbolic tangent activation function is also referred to simply as the Tanh (also “tanh” and “TanH”) function. It is very similar to the sigmoid activation function and even has the same S-shape.

SGD: Stochastic Gradient Descent, or SGD for short, is an optimization algorithm used to train machine learning algorithms, most notably artificial neural networks used in deep learning. Optimization is a type of searching process and this algorithm is called “gradient descent“, where “gradient” refers to the calculation of an error gradient or slope of error and “descent” refers to the moving down along that slope towards some minimum level of error. The algorithm is iterative. This means that the search process occurs over multiple discrete steps, each step hopefully slightly improving the model parameters.

Adam: Adam is a replacement optimization algorithm for stochastic gradient descent for training deep learning models. Adam combines the best properties of the AdaGrad and RMSProp algorithms to provide an optimization algorithm that can handle sparse gradients on noisy problems.

CrossEntropy Loss: Cross-entropy loss is used when adjusting model weights during training. The aim is to minimize the loss, i.e, the smaller the loss the better the model. A perfect model has a cross-entropy loss of 0.

❖ Model Diagram:

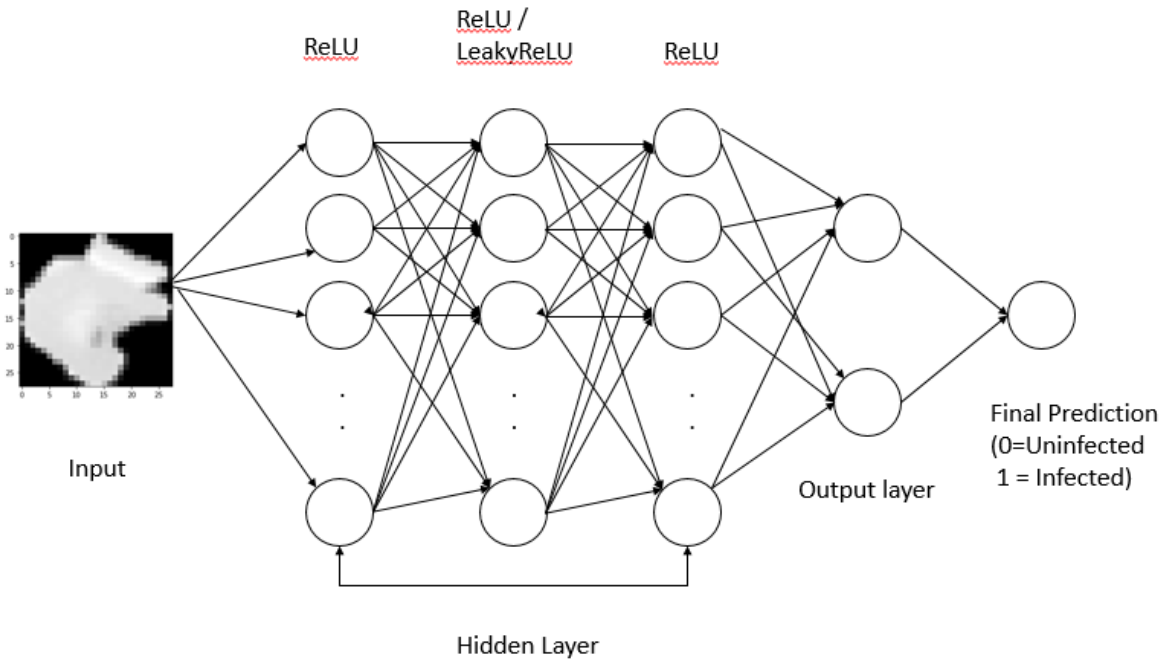


Figure: Diagram of Malaria Detection Model.

Experiments:

- ❖ **Dataset:** Malaria is a public dataset which we collected from kaggle. This dataset consists of 27446 RGB samples divided into two classes named: Parasitized and uninfected. Parasitized class has 13779 samples and Uninfected class has 13667 samples.

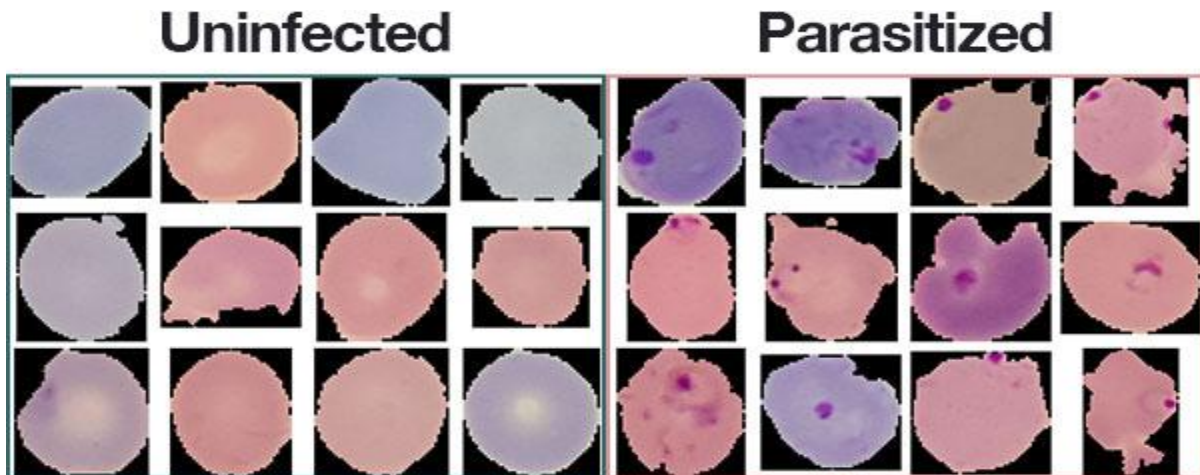


Figure: Snapshot of the Dataset.

- **Train-Test Splitting Ratio:** The entire dataset is splitted with proportion to 80:20 and 70:30. Data are assigned randomly in the training, validation and testing set.

Ratio	Training Data	validation Data	Testing
(80:20)	21957	2195	5489
(70:30)	19213	1921	8233

❖ Evaluation Metric:

- **Accuracy:** Accuracy is the most popular performance matrix which measures how often the classifier produces the correct prediction. It is easily suited for binary classification along with multiclass classification problems.

$$\text{Accuracy} = (TP + TN) / (TP + FP + FN + TN)$$

= correct predictions / Total samples.

In other words it can be defined as the ratio of the number of correct predicted images and the total number of images.

- **Training accuracy:** The accuracy of a model on examples it was constructed on.
- **Testing accuracy:** The test accuracy is the accuracy of a model on examples it hasn't seen.
- **Confusion matrix:** A tabulation of the predicted class (usually vertically) against the actual class (thus horizontally).
- **Precision:** Precision is a valid choice of evaluation metric when we want to be very sure of our prediction.
- **Recall:** Recall is a valid choice of evaluation metric when we want to capture as many positives as possible.

❖ Results:

- **Setting 1:**
 - batch_size = 64
 - num_iters = 30000
 - num_hidden = 100
 - learning_rate = 0.1
 - Accuracy = 74.94
 - Activation function = Relu, Leaky relu.

Optimizer = SGD

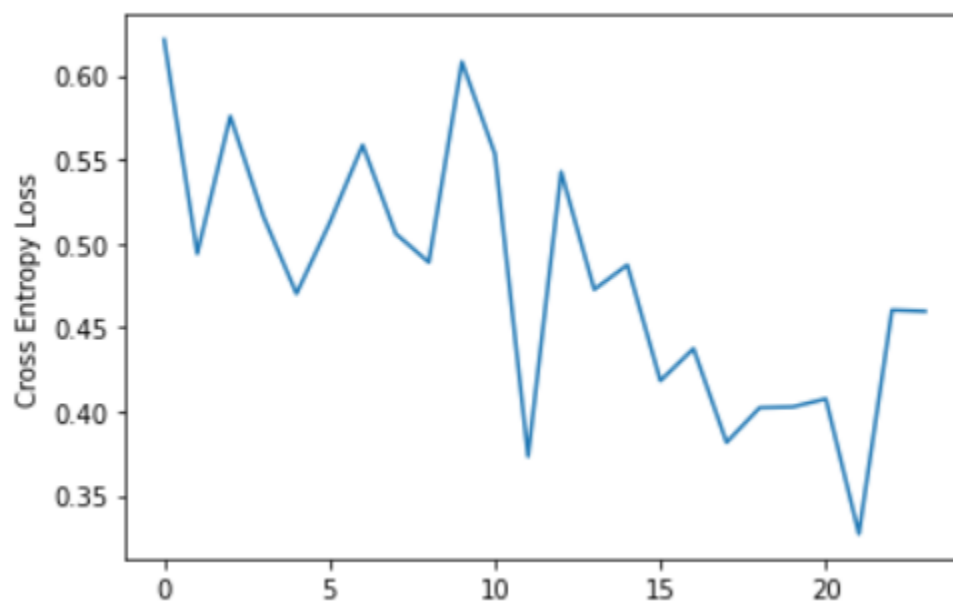


Figure: Loss Graph

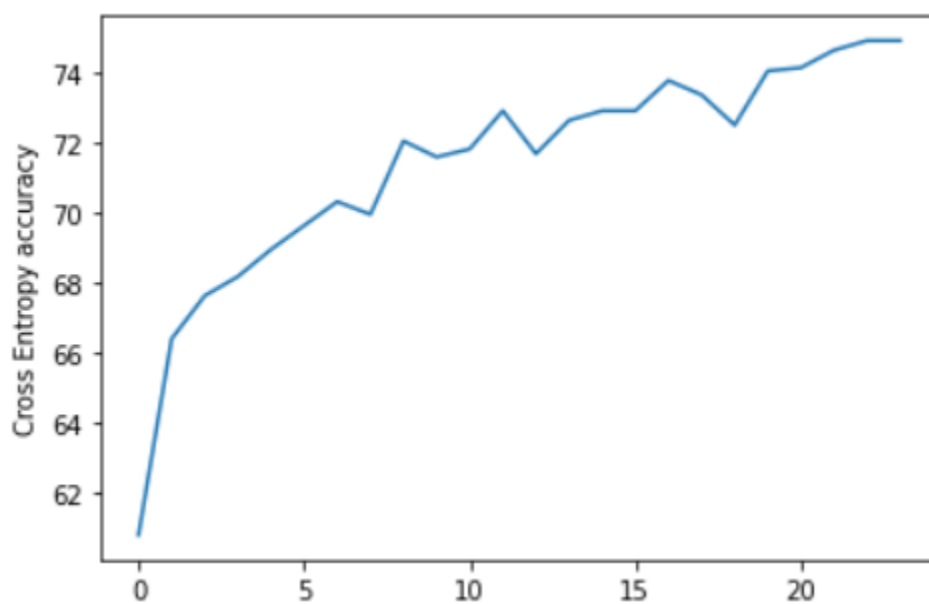


Figure: Accuracy Graph

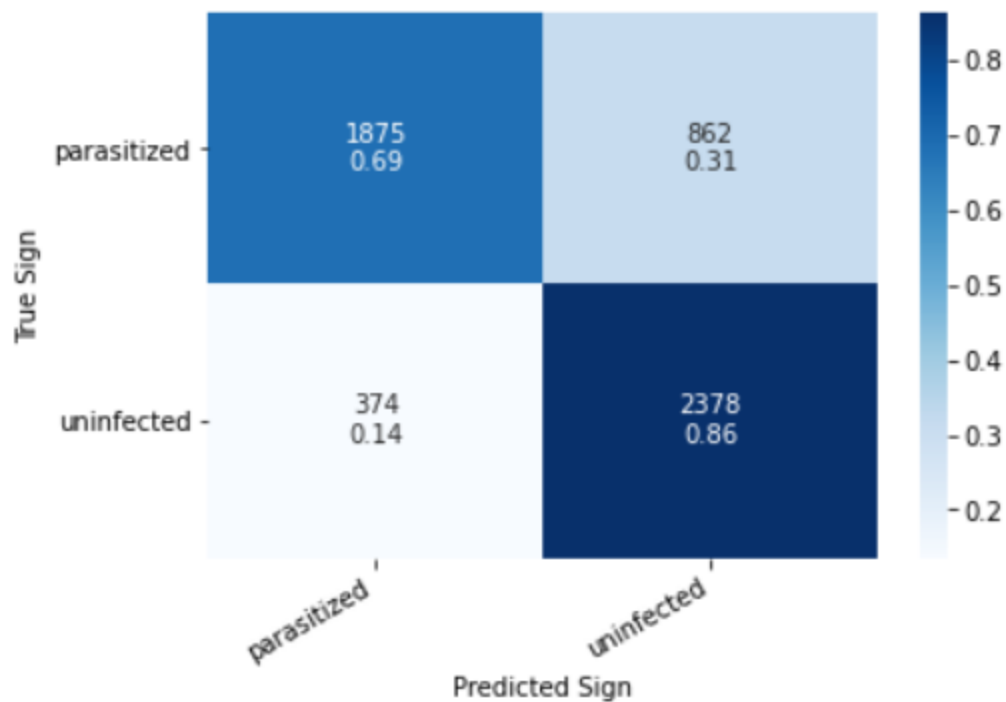


Figure: Confusion Matrix

	precision	recall	f1-score	support
parasitized	0.83	0.69	0.75	2737
uninfected	0.73	0.86	0.79	2752
accuracy			0.77	5489
macro avg	0.78	0.77	0.77	5489
weighted avg	0.78	0.77	0.77	5489

Figure: Classification Report

➤ **Setting 2:**

batch_size = 64

num_iters = 15000

num_hidden = 1024, 560, 256

learning_rate = 0.0001

Accuracy = 77.17

Activation function = Relu
Optimizer = Adam

Accuracy 77.17 is gained after using relu as an activation function and Adam as an optimizer where Leaky relu and relu both were used as an activation function in settings 1. We have also changed the number of neurons in settings 2(1024, 560, 256) where the number of neurons was 100 in both three layers. Number of iterations and learning rate are also reduced in settings 2.

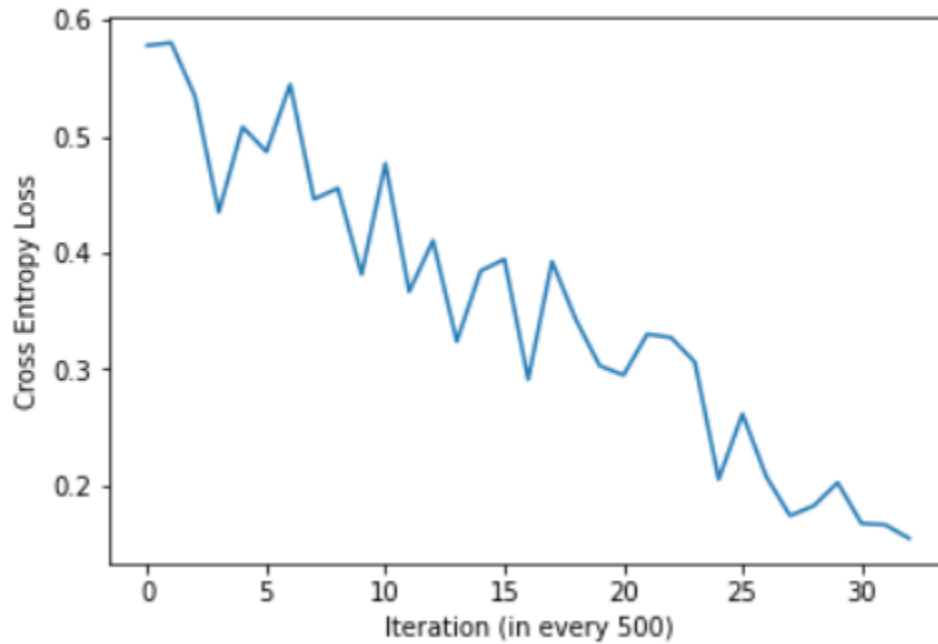


Figure: Loss Graph

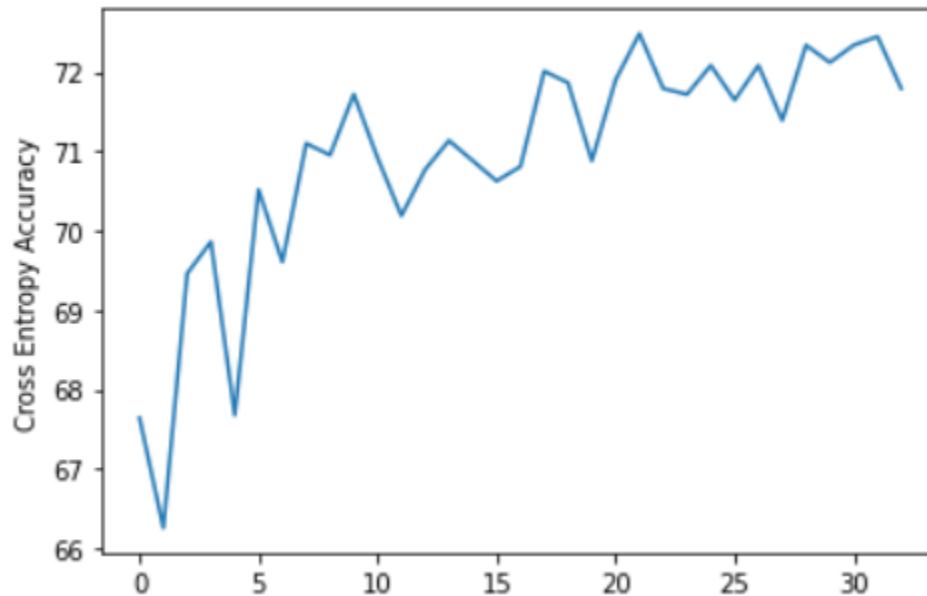


Figure: Accuracy Graph

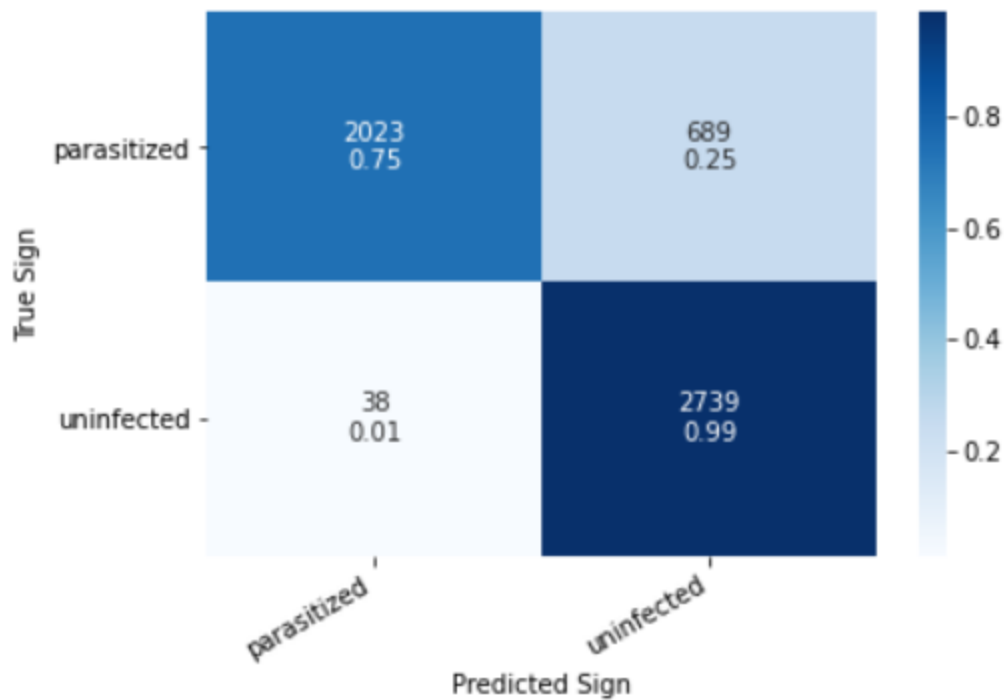


Figure: Confusion Matrix

	precision	recall	f1-score	support
parasitized	0.98	0.75	0.85	2712
uninfected	0.80	0.99	0.88	2777
accuracy			0.87	5489
macro avg	0.89	0.87	0.87	5489
weighted avg	0.89	0.87	0.87	5489

Figure: Classification Report

➤ **Setting 3:**
 batch_size = 64
 num_iters = 21000
 num_hidden = 1000, 600, 200
 learning_rate = 0.0001
 Accuracy = 71.79
 Activation function = Relu, Tanh.
 Optimizer = Adam

Accuracy 71.79 is gained after using relu and tanh as an activation function where relu was used as an activation function in settings 2. We have also changed the number of neurons in settings 3(1000, 600, 200) where the number of neurons were 1024, 560, 256 . Number of iterations is increased in settings 3.

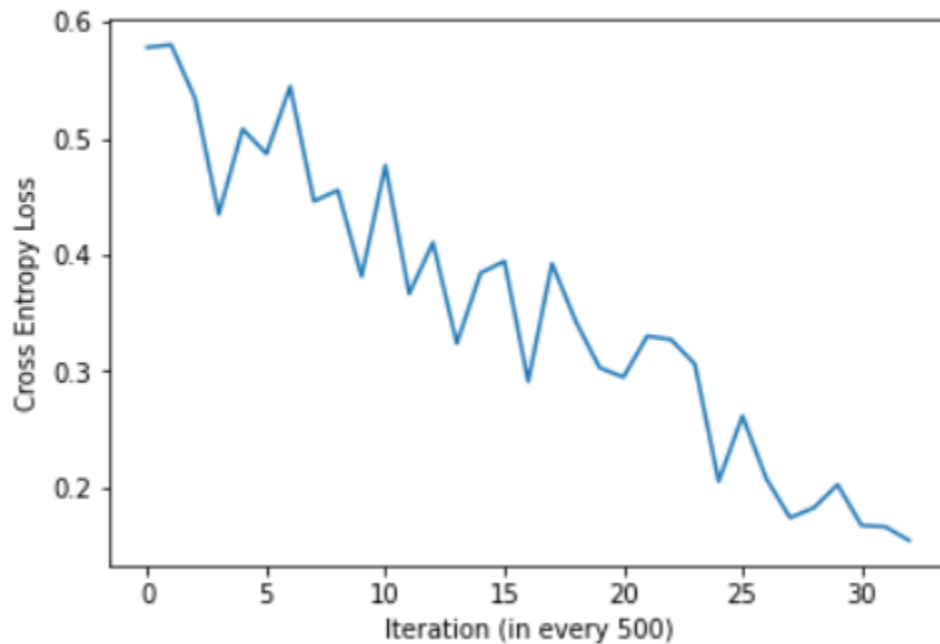


Figure: Loss Graph

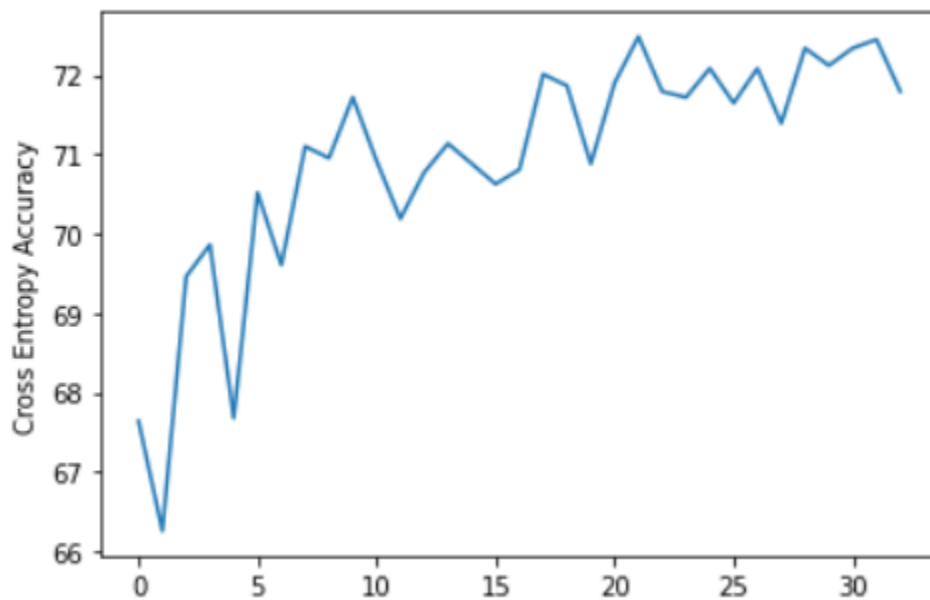


Figure: Accuracy Graph

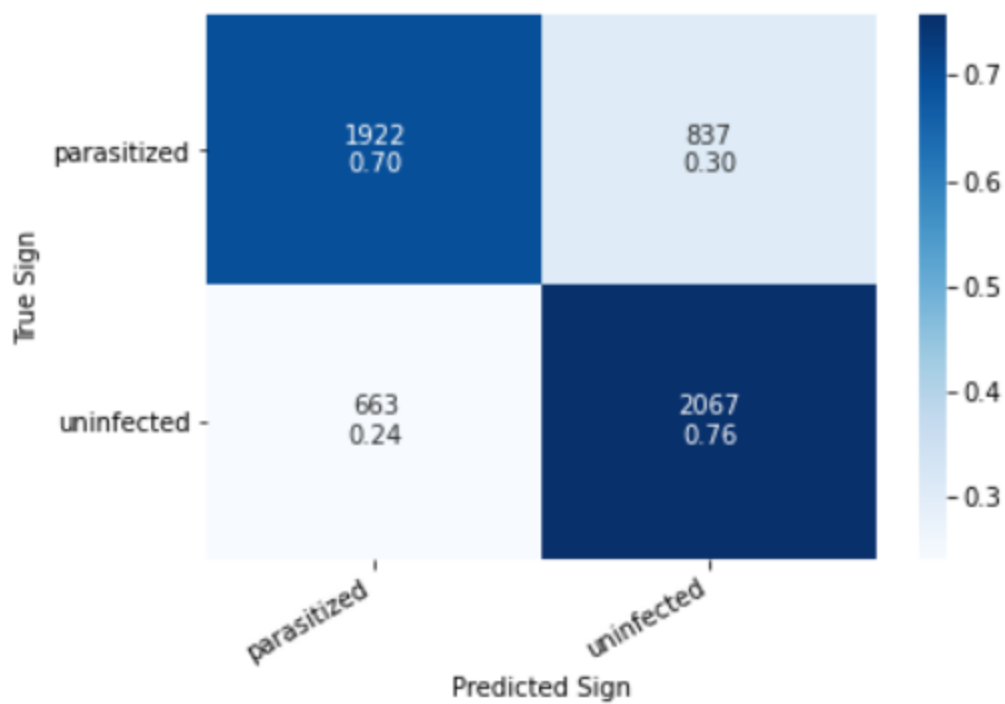


Figure: Confusion Matrix

	precision	recall	f1-score	support
parasitized	0.74	0.70	0.72	2759
uninfected	0.71	0.76	0.73	2730
accuracy			0.73	5489
macro avg	0.73	0.73	0.73	5489
weighted avg	0.73	0.73	0.73	5489

Figure: Classification Report

	80:20		70:30	
Models	Validation	Testing	Validation	Testing
Model 1	74.94	82.81	76.73	70.31
Model 2	77.17	89.20	74.67	71.88
Model 3	71.79	81.50	75.32	84.38

Conclusion:

In the proposed system, we used deep neural networks to detect malaria disease. This system is tested for the detection of the Malaria parasites in blood samples. The image dataset was splitted into 80:20. We tried 3 settings using different batch size, iteration number, number of neurons, learning rate, activation function, optimizer and we got different accuracy. The highest validation accuracy was 77.17% and testing accuracy was 89.20%. The accuracy would have been much better if Convolutional Neural Network was used instead of Deep Neural Network.